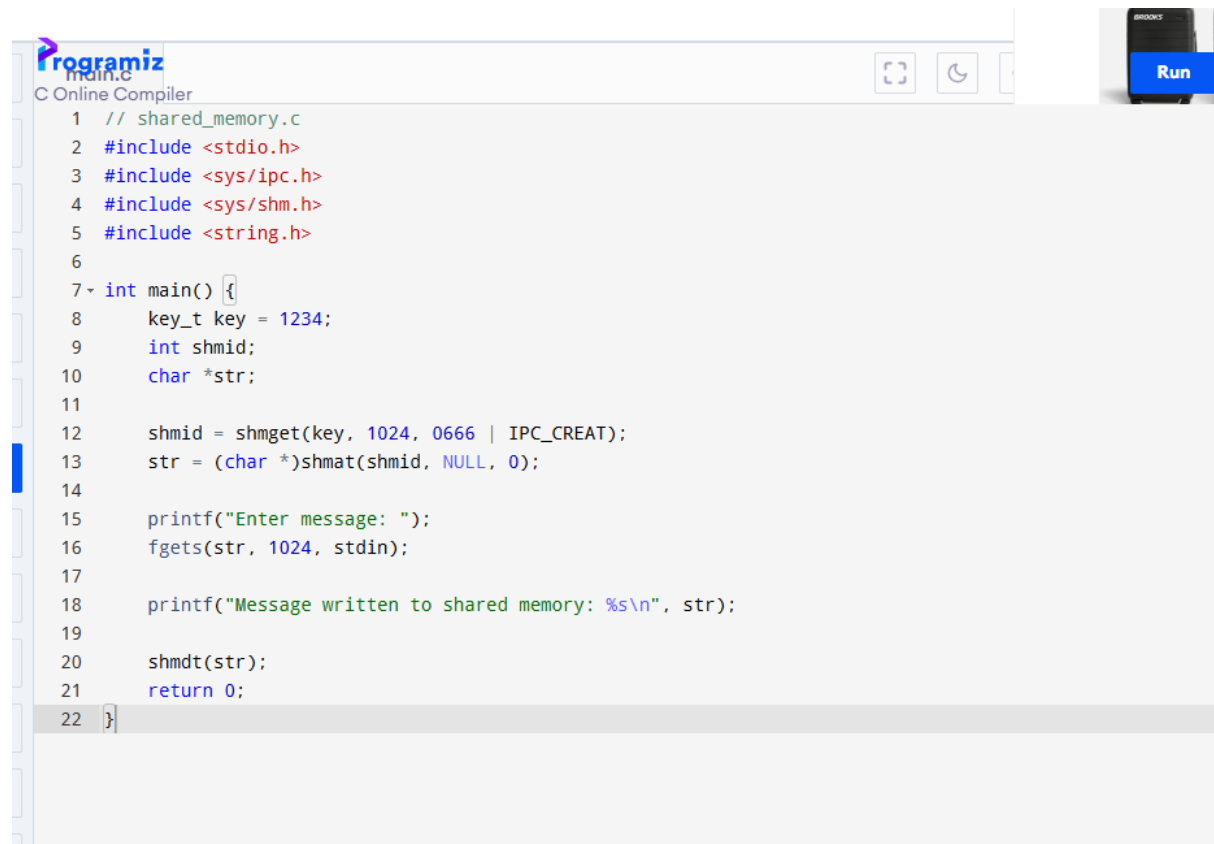


9. Illustrate the concept of inter-process communication using shared memory with a C program.

Code:

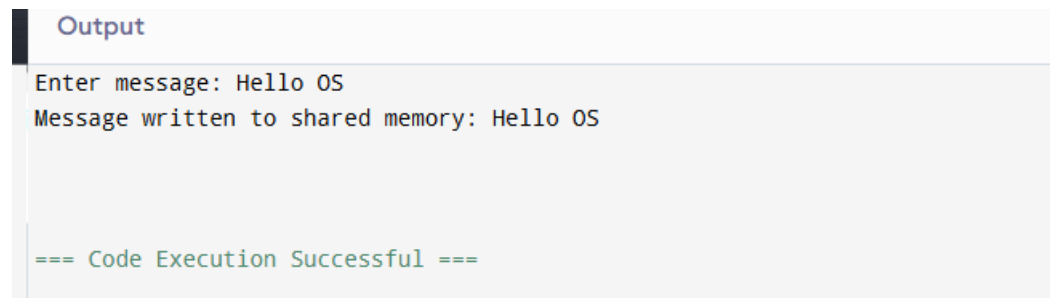


The screenshot shows a web-based C compiler interface. The title bar says 'Programiz' and 'main.c'. Below it, it says 'C Online Compiler'. The code is as follows:

```
1 // shared_memory.c
2 #include <stdio.h>
3 #include <sys/ipc.h>
4 #include <sys/shm.h>
5 #include <string.h>
6
7 int main() {
8     key_t key = 1234;
9     int shmid;
10    char *str;
11
12    shmid = shmget(key, 1024, 0666 | IPC_CREAT);
13    str = (char *)shmat(shmid, NULL, 0);
14
15    printf("Enter message: ");
16    fgets(str, 1024, stdin);
17
18    printf("Message written to shared memory: %s\n", str);
19
20    shmdt(str);
21    return 0;
22 }
```

On the right side of the editor, there are icons for full screen, settings, and a 'Run' button.

Output:



The screenshot shows the output of the program. It has a title bar 'Output'. The output text is:

```
Enter message: Hello OS
Message written to shared memory: Hello OS

=== Code Execution Successful ===
```